POWER PROFILE SYSTEM

Electrical Power System Design & Analysis Tool

Comprehensive Technical Report

Equations · Standards · Algorithms · Architecture

Project Type:	Graduation Project — Electrical Engineering
Tech Stack:	Laravel 13 / PHP 8.4 + React 18 / Vite
Standards:	IEC 60364-8-1 · BS 7671 · PENRA · NEC · CIBSE
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Date:	June 2026

Table of Contents

1. Project Overview & Motivation

Power Profile is a full-stack web application designed and developed as a graduation project in electrical engineering. Its primary purpose is to automate the complex, multi-step process of designing an electrical power supply system for multi-building complexes — a task that traditionally requires specialist knowledge and many hours of manual calculation.

The tool takes as input a hierarchical description of a site (project → buildings → floors → rooms → electrical components) and produces as output:

- Total demand (VA, kW, kVAR) with diversity factors applied per IEC 60364-8-1
- Power factor correction requirement and capacitor bank sizing
- 24-hour load profile derived from usage schedules
- Solar PV generation profile using real NASA POWER satellite irradiance data
- Battery Energy Storage System (BESS) state-of-charge simulation
- Multi-source dispatch (Solar → Battery → Grid → Generator) hour-by-hour
- 3-phase load balancing with optimal phase assignment
- PDF and structured backup/restore reports

The engineering calculations are grounded in internationally accepted standards including **IEC 60364-8-1** (energy efficiency in buildings), **BS 7671** (UK wiring regulations), **CIBSE Guide C** (building services engineering), **PENRA** (Power & Energy Regulatory Authority power-factor targets), and **NEC Article 430** (motor branch circuit sizing). The solar model implements Spencer's declination formula and the classical hour-angle sunrise/sunset algorithm.

1.1 Technology Stack

Layer	Technology	Version	Role
Backend	Laravel / PHP	13 / 8.4	REST API, business logic, auth
Frontend	React + Vite	18 / 4.x	Single-page application
Styling	Tailwind CSS	3.x	Utility-first responsive UI
Charts	Recharts	2.x	24-h load & dispatch visualisation
Database	SQLite	3.x	Relational data store
Auth	Laravel Sanctum + Google OAuth	2.0	Token-based API auth
Solar data	NASA POWER API	v2	Satellite GHI irradiance
Routing	React Router	v6	Client-side navigation

2. System Architecture

The application follows a classic client-server architecture with a clear four-layer separation of concerns. The React SPA communicates with a Laravel REST API exclusively via JSON over HTTP. All engineering calculations are performed server-side inside dedicated Service classes, keeping the frontend entirely presentation-focused.

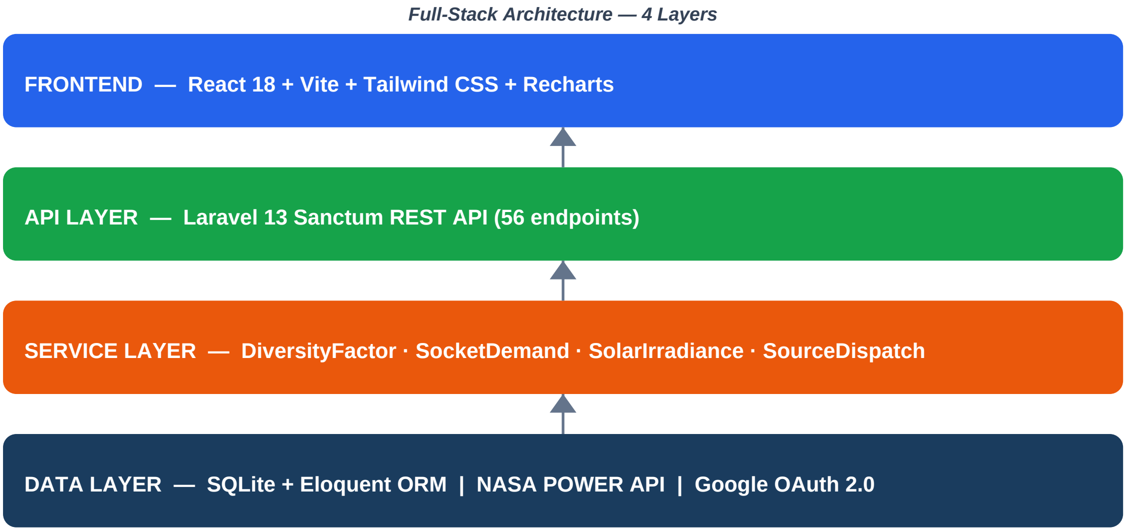


Figure 2.1 — Four-layer full-stack architecture

2.1 Request Lifecycle

- User interacts with a React page (e.g. opens Load Schedule).
- React component calls axios GET /api/projects/{id}/load-profile.
- Laravel route resolves to LoadProfileController@show.
- Controller calls DiversityFactorService, SocketDemandService, and SolarIrradianceService.
- SolarIrradianceService queries NASA POWER (or returns cached data).
- Controller aggregates all component data into 24-hour hourly arrays.
- JSON response returned; React stores in state and renders Recharts graphs.

2.2 Project Hierarchy (Data Model Summary)

Every entity in the system forms a strict containment tree. Electrical components (loads) can be attached at any level of this tree. Diversity factors are then applied top-down according to which level the component sits at.

Level	Entity	Multiplicity	Electrical Meaning
0	Project	1 per user	Entire site / campus
1	Building	1–N per project	Individual structure

Level	Entity	Multiplicity	Electrical Meaning
2	Floor	1–N per building	Storey
3	Room	1–N per floor	Space / zone
4	Component	1–N per room/floor/building/project	Individual load

3. Electrical Standards & Theory

The calculations implemented in this project draw on the following internationally recognised standards and references.

Standard / Reference	Full Title	Application in This System
IEC 60364-8-1	Low-voltage electrical installations — Energy efficiency	Diversity factors by building type
BS 7671:2018	Requirements for Electrical Installations (IET Wiring Regs)	Diversity factor source tables
CIBSE Guide C	Reference Data — Building Services Engineering	Peak coincidence factors
PENRA	Power and Energy Regulatory Authority — PF regulation	Target power factor = 0.95
NEC Article 430	National Electrical Code — Motors	125% motor inrush rule
IEC 60947-4	Low-voltage switchgear — Contactors and starters	Motor inrush standard
IEC 61675-3	Solar radiation — Hour-angle and declination	Sunrise/sunset computation
Spencer (1971)	Fourier series for solar declination	Day-of-year declination formula
NASA POWER v2	Prediction of Worldwide Energy Resources	Satellite GHI irradiance data
IEC 60831	Shunt power capacitors — General performance	Capacitor bank configuration

4. Diversity Factor System (IEC 60364-8-1)

In any real building, not all electrical loads operate simultaneously at full rated capacity. The **diversity factor** (DF) accounts for this statistical reality by scaling the sum of individual rated loads to a realistic maximum demand. IEC 60364-8-1 and BS 7671 provide DF tables by building type.

4.1 Mathematical Definition

The general form of the diversity-weighted demand at a given aggregation level is:

```
P_demand = Σ■ ( P_rated,■ × DF_building × DF_room_type × DF_coincidence )
```

where:

- P_rated,■ = rated apparent power of component i [VA]
- DF_building = tabulated factor for the building type (room-to-floor leg)
- DF_room_type = coincidence factor for the specific room type (0.25 – 1.00)
- DF_coincidence = 0.70 (project-level top-of-hierarchy factor, IEC 60364-8-1 §8.3)

4.2 Hierarchical DF Application

The system applies diversity factors in a strict hierarchy. A component defined at a *room* level receives the full three-layer diversification; one defined at *project* level receives none (it is already at the top).

```
Room component: DF = DF_room_type × DF_floor_to_bldg × DF_room_to_floor × 0.70 Floor
component: DF = DF_floor_to_bldg × 0.70 Building component: DF = 0.70 Project
component: DF = 1.00 (already at system level) CRITICAL priority: DF = 1.00 (never
diversified — always on)
```

4.3 Building-Type Diversity Factors

Building Type	Room → Floor DF	Floor → Building DF
Residential House	0.60	0.70
Residential Apartment	0.65	0.70
Hotel	0.65	0.70
Office	0.85	0.80
Educational School	0.80	0.80
Educational University	0.85	0.80
Retail	0.85	0.85
Hospital	0.90	0.90
Industrial	0.85	0.85
Mosque / Worship	0.80	0.75
Sports	0.80	0.80

Building Type	Room → Floor DF	Floor → Building DF
Generic / Default (IEC)	0.90	0.80

■ *Source: IEC 60364-8-1:2019, Table B.1 / BS 7671 Appendix 1 / CIBSE Guide C Table 14.4*

4.4 Room-Type Coincidence Factors

Within each building type, individual room types receive an additional coincidence factor reflecting typical simultaneous usage patterns.

Room Type	Coincidence Factor	Rationale
Server Room	1.00	Continuous operation — no diversity
Operating Theatre	1.00	Life-safety — no diversity
Laboratory	0.90	High utilisation, scheduled use
Classroom / Lecture	0.85	Group use during sessions
Workshop / Retail	0.85	High occupancy during open hours
Open Office / Gym	0.80	Moderate simultaneous use
Commercial Kitchen	0.75	Staggered meal preparation
Private Office / Prayer Hall	0.75	Individual or scheduled use
Reception / Meeting	0.70	Variable occupancy
Corridor / Living	0.60	Transient occupancy
Residential Kitchen	0.55	Short peak cooking periods
Hotel Room	0.50	Guests absent much of the time
Bedroom	0.45	Night-time use only
Warehouse / Storage	0.30	Infrequent access
Bathroom	0.25	Very short occupation periods
Default	0.80	IEC generic building default

5. Socket / Outlet Demand Calculations

Electrical socket outlets contribute a significant portion of building demand but are rarely all loaded simultaneously. The system implements a tiered demand factor schedule consistent with IEC 60364-5-52 and general industry practice.

5.1 Rated VA per Outlet

```
VA_per_outlet = 200 VA (IEC standard residential/commercial outlet rating)
```

5.2 Tiered Demand Factor

For n outlets at a single point-of-supply, the demand is calculated as:

```
If n ≤ 10: Demand = n × 200 × 1.00 If 10 < n ≤ 20: Demand = 10×200×1.00 + (n-10)×200×0.75 If n > 20: Demand = 10×200×1.00 + 10×200×0.75 + (n-20)×200×0.40 General form: Demand = min(n,10)×200 + min(max(n-10,0),10)×150 + max(n-20,0)×80
```

Outlet Range	Unit Demand	Demand Factor	Reasoning
1st – 10th outlets	200 VA each	100%	All likely in simultaneous use
11th – 20th outlets	150 VA each	75%	Moderate probability of use
21st outlet onwards	80 VA each	40%	Statistically low probability

5.3 System-Size Coincidence Factor

When aggregating socket demand from many rooms, an additional coincidence factor is applied based on total system size:

System Total Demand	Coincidence Factor
< 50 kVA	1.00
50 kVA – 250 kVA	0.92
250 kVA – 1000 kVA	0.85

5.4 Hierarchical Socket Aggregation

```
Room demand = tiered_demand(n_room_sockets) Floor demand = tiered_demand(Σ room sockets + floor own sockets) Bldg demand = tiered_demand(all floors) × coincidence_factor(system_kva) Project demand= tiered_demand(all buildings) × coincidence_factor(system_kva)
```

6. Active, Reactive & Apparent Power Theory

All AC electrical loads are characterised by three distinct power quantities. Understanding their relationships is fundamental to sizing cables, transformers, generators, and power-factor correction equipment.

6.1 The Power Triangle

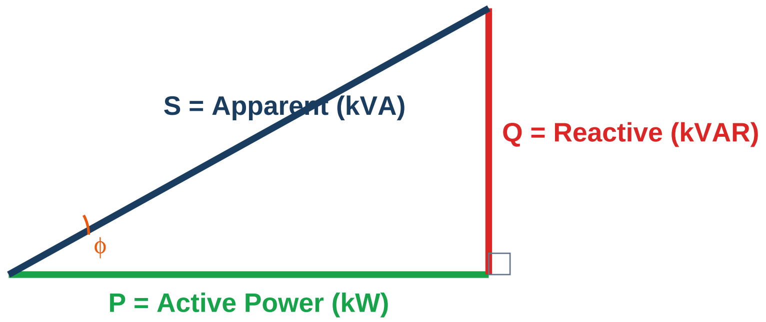


Figure 6.1 — Power triangle: P (active), Q (reactive), S (apparent)

6.2 Core Power Equations

Quantity	Formula
Apparent Power	$S\ [VA] = \sqrt{(P^2 + Q^2)}$
Active Power	$P\ [W] = S \times \cos(\phi)$
Reactive Power	$Q\ [VAR] = S \times \sin(\phi) = P \times \tan(\phi)$
Power Factor	$PF = \cos(\phi) = P / S$
Phase Angle	$\phi = \arccos(PF)$
Q from PF	$Q = P \times \tan(\arccos(PF))$
3-phase apparent	$S_{3\phi} = \sqrt{3} \times V_{LL} \times I_L\ [V_{LL} = 400\ V]$

Quantity	Formula
1-phase apparent	$S_{1\phi} = V_{LN} \times I_L$ [$V_{LN} = 230$ V]
3-phase current	$I_L = S / (\sqrt{3} \times 400)$ [A]

6.3 Per-Component Reactive Power

For each electrical component i with rated apparent power S_i and power factor PF_i:

```
P_i = S_i × PF_i ϕ_i = arccos(PF_i) Q_i = P_i × tan(ϕ_i) = S_i × sin( arccos(PF_i) ) S_i
= √(P_i² + Q_i²) ✓ (consistency check)
```

6.4 System Aggregation (Vector Sum)

Individual component phasors are summed as complex numbers before computing the resultant power factor — scalar addition of VA values would overestimate S if loads have different power factors.

```
P_total = Σ P_i (scalars — always additive) Q_total = Σ Q_i (scalars for DF-weighted
inductive loads) S_total = √( P_total² + Q_total² ) PF_total = P_total / S_total
I_total = S_total / (√3 × 400) [3-phase]
```

7. Power Factor Correction & Capacitor Sizing

When the system power factor falls below the PENRA-mandated threshold of 0.85, shunt capacitor banks must be installed to inject leading reactive power (VAR) and raise the effective PF toward the target of 0.95. This reduces billing penalties, decreases line currents, and lowers I^2R losses.

7.1 Reactive Power Deficit

```
ϕ_current = arccos( PF_current ) [current phase angle] ϕ_target = arccos( 0.95 ) ≈
18.19° [PENRA target] Q_current = P × tan( ϕ_current ) [kVAR absorbed by loads]
Q_target = P × tan( ϕ_target ) [kVAR at PF = 0.95] Q_cap_needed = Q_current − Q_target
[kVAR to inject]
```

7.2 Capacitor Bank Sizing (Step Rounding)

Capacitor banks are manufactured in standard discrete steps of 0.5 kVAR. The required bank is rounded up to the nearest 0.5 kVAR step.

```
Q_bank = ■ Q_cap_needed / 0.5 ■ × 0.5 [kVAR]
```

7.3 Per-Phase Capacitance — Delta (Δ) Configuration

The capacitors are connected in delta (Δ) across the 400 V 3-phase supply. In delta connection each capacitor sees the full line-to-line voltage of 400 V. Total reactive injection Q_bank is equally split among three phases.

```
Q_per_phase = Q_bank × 1000 / 3 [VAR per capacitor] For a capacitor across 400 V at 50
Hz: I_cap = Q_per_phase / V_LL X_c = V_LL / I_cap = V_LL² / Q_per_phase X_c = 1 / (2π f
C) Therefore: C = Q_per_phase / (2π f V_LL²) [Farads] C = (Q_bank×1000/3) / (2 × π × 50
× 400²) [Farads] C_μF = C × 1×10■ [microfarads]
```

■ *PENRA threshold: PF < 0.85 triggers mandatory correction. Target: PF = 0.95.*

■ *Capacitor step: 0.5 kVAR (standard manufactured increment, IEC 60831).*

■ *Delta connection chosen because it provides the same reactive power as star (Y) with 1/3 the capacitance per unit, reducing component count.*

7.4 Correction Thresholds

Power Factor Range	Status	Action
PF ≥ 0.95	Compliant	No correction needed
0.85 ≤ PF < 0.95	Warning	Improvement recommended
PF < 0.85	Non-compliant	Capacitor bank mandatory (PENRA)

8. Motor Inrush Current (NEC Article 430 / IEC 60947-4)

When an AC induction motor starts, it draws a starting (inrush) current typically 5–7× its full-load current for a duration of 0.5–10 seconds. NEC Article 430 and IEC 60947-4 both require that distribution boards and protective devices be sized for 125% of the largest motor's full-load current to ensure safe operation and prevent nuisance tripping.

8.1 NEC 125% Rule

```
Identify: Motor_max_VA = max( VA_i ) for all motor components Inrush VA: VA_inrush =
Motor_max_VA × 1.25 Addition: ΔVA = VA_inrush − Motor_max_VA = 0.25 × Motor_max_VA
Apply to: S_max (peak apparent power vector) Q_max (peak reactive power vector) NOT to:
S_optimized (diversity-weighted demand)
```

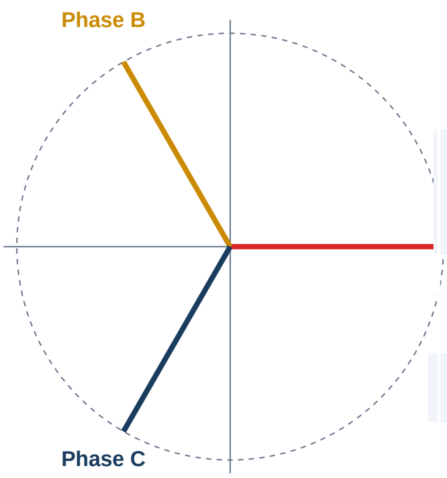
The 25% addition is applied only to the *maximum demand* vector used for short-circuit and cable sizing, not to the optimised (diversified) demand used for energy calculations. This correctly sizes protective devices without inflating the energy billing estimate.

8.2 Motor Components in the Database

Component types are flagged with **is_motor = true** in the component type library. The system automatically identifies the largest motor VA in the project hierarchy at calculation time, so no manual intervention is required.

9. Three-Phase Load Balancing

In a 3-phase 4-wire system (400 V L-L / 230 V L-N, 50 Hz), single-phase loads must be distributed across phases A, B, and C to minimise neutral current and reduce transformer losses. Excessive imbalance causes increased I^2R losses, elevated neutral current, and potential transformer overheating.



3-Phase Phasor (120° separation)

9.1 Phase Phasor Representation

Each phase in a balanced 3-phase system is displaced by 120°:

```
V_A = V_phase ∠ 0° V_B = V_phase ∠ 120° V_C = V_phase ∠ 240° V_phase = 230 V (line-to-neutral) V_LL = V_phase × √3 = 400 V
```

The current on each phase from a single-phase load of apparent power S:

```
I_phase = S / V_LN = S / 230 [A]
```

9.2 Phasor Aggregation per Phase

```
For each phase X ∈ {A, B, C}: I_X_phasor = Σ ( S_i / 230 ) ∠ angle_X for all 1-phase loads on phase X where angle_A = 0°, angle_B = 120°, angle_C = 240° |I_X| = magnitude of the phasor sum for phase X Imbalance % = ( max|I| − min|I| ) / avg|I| × 100
```

9.3 Imbalance Thresholds

Imbalance %	Severity	Action
< 10%	Acceptable	No action required
10–20%	Warning	Redistribution recommended
> 20%	Critical	Immediate rebalancing required

9.4 Greedy Optimal Phase Assignment Algorithm

When the user triggers 'Apply Optimal Phase Assignment', the backend executes a greedy algorithm that assigns each unphased 1-phase component to the currently most lightly-loaded phase:

```
Sort components by VA descending (largest loads first) For each component i in sorted order: X* = argmin( |I_A|, |I_B|, |I_C| ) ← phase with lowest current Assign component i → phase X* Update |I_X*| += S_i / 230
```

This greedy approach yields a near-optimal distribution in O(n log n) time. The 'largest first' ordering (a variant of First-Fit Decreasing bin-packing) has been shown to produce solutions within 11/9 of optimal in the worst case.

10. Solar PV Modelling & NASA POWER Integration

The system generates a realistic hourly solar generation profile for any geographic coordinate by combining NASA's satellite-derived Global Horizontal Irradiance (GHI) data with an analytical sinusoidal bell-curve model of daily irradiance variation.

10.1 Available Roof Area & PV Capacity

In practice, not all roof area is usable for PV panels due to obstructions, setbacks, maintenance walkways, and shading. A coverage ratio of 17% is used:

```
Usable_area = Roof_area_m² × 0.17 Capacity_W = Usable_area × STC_irradiance × PR_sizing
= Roof_area × 0.17 × 1000 W/m² × 0.75 where: STC_irradiance = 1000 W/m² (Standard Test
Conditions) PR_sizing = 0.75 (conservative performance ratio for sizing) PR_energy =
0.80 (performance ratio used in energy calculations)
```

■ *The performance ratio accounts for inverter losses, temperature derating, soiling, shading, and cable losses. Typical values: 0.75–0.85.*

10.2 Solar Declination (Spencer's Equation, 1971)

The solar declination δ is the angle between the equatorial plane and the line from Earth's centre to the Sun. It varies from +23.45° (summer solstice) to –23.45° (winter solstice).

```
doy = day-of-year of the mid-month (e.g., January 15 → doy = 15) δ = 23.45 × sin(
360/365 × (doy − 81) ) [degrees] (Spencer 1971 simplified — full Fourier has 6 terms;
this is ±0.3°)
```

10.3 Sunrise / Sunset — Hour-Angle Formula

The hour angle H at sunrise/sunset satisfies $\cos(H) = -\tan(\phi)\cdot\tan(\delta)$, where ϕ is the site latitude and δ is the declination.

```
cos(H_ss) = −tan(ϕ_lat) × tan(δ) Special cases: cos(H_ss) > +1.0 → Polar night (sun
never rises) cos(H_ss) < −1.0 → Midnight sun (sun never sets) Otherwise: H_ss_deg =
arccos( cos(H_ss) ) [degrees] Sunrise = 12:00 − H_ss_deg / 15 [hours local solar time]
Sunset = 12:00 + H_ss_deg / 15 [hours local solar time] Daylight = Sunset − Sunrise
[hours] Note: divide by 15 because Earth rotates 15°/hour
```

10.4 Hourly Generation Profile (Sinusoidal Bell Curve)

The actual irradiance during daylight hours follows approximately a sinusoidal shape. The normalised daily energy equals PSH × PR:

```
Peak_W = Capacity_W × PSH_kWh/m²/day × PR_energy × π / ( 2 × Daylight_hours ) [This
ensures ∫■²■ Output(t) dt = Capacity × PSH × PR (energy conservation)] For each hour h
(0 – 23): mid = h + 0.5 (mid-point of hour) if mid ≤ Sunrise or mid ≥ Sunset: Output[h]
= 0 W else: t = (mid − Sunrise) / Daylight ∈ [0, 1] Output[h] = Peak_W × sin(π × t)
```

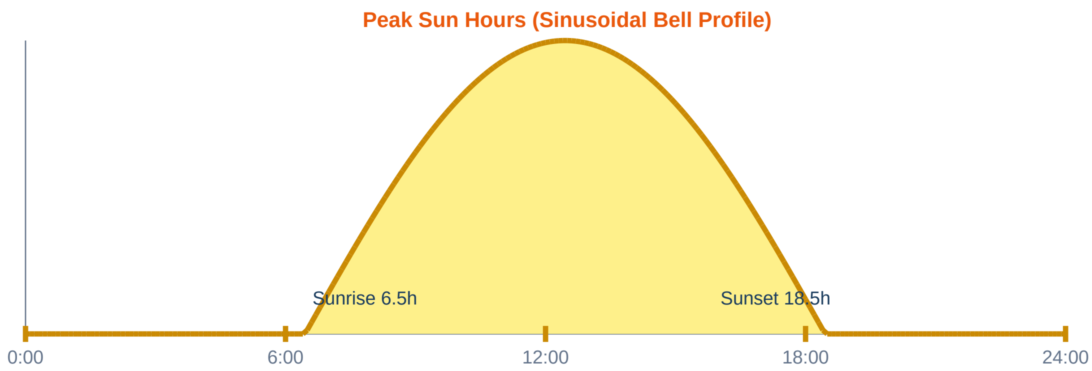


Figure 10.1 — Sinusoidal daily solar generation profile (illustrative, latitude 30°N, June)

10.5 Peak Sun Hours Lookup Table

The PSH table provides monthly average daily GHI (kWh/m²/day) at seven latitude bands. Values for intermediate latitudes are linearly interpolated.

Lat\Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
0°	5.5	5.8	6.0	5.9	5.6	5.5	5.5	5.7	5.9	5.8	5.5	5.4
10°	5.0	5.5	6.0	6.2	6.1	6.0	6.0	6.1	6.1	5.8	5.2	4.8
20°	4.5	5.2	5.9	6.4	6.5	6.4	6.4	6.4	6.1	5.6	4.8	4.3
30°	3.8	4.6	5.6	6.4	6.8	7.0	6.9	6.6	6.1	5.2	4.1	3.5
40°	2.9	3.8	5.0	6.1	6.9	7.3	7.1	6.5	5.6	4.4	3.2	2.5
50°	1.8	2.9	4.3	5.7	6.7	7.2	7.0	6.1	4.9	3.4	2.1	1.4
60°	0.8	1.8	3.4	5.1	6.4	7.0	6.8	5.6	4.0	2.4	1.1	0.5

■ *Southern hemisphere: season flipped by 6 months (July ↔ January, etc.)*

10.6 NASA POWER API Integration

When a project has GPS coordinates set, the system queries the NASA POWER REST API for real satellite-measured GHI at the representative day of each month (15th). This is more accurate than the static lookup table.

Parameter	Value
API Base URL	https://power.larc.nasa.gov/api/temporal/hourly/point
Parameter Code	ALLSKY_SFC_SW_DWN (All-sky surface shortwave downward irradiance)
Temporal	Hourly, representative day (15th of month)
Units	W/m²
Accuracy	±3% for monthly means
Cache Duration	30 days (historical satellite data is stable)
Fallback	Static PSH lookup table if API unavailable or coordinates absent

11. Battery Energy Storage System (BESS)

The system models a battery bank of configurable chemistry, size, and age. Five battery chemistries are pre-configured with datasheet-accurate parameters. Capacity degrades with age and cycle count, modelled through an age factor.

11.1 Supported Battery Chemistries

Chemistry	DoD	Round-trip η	Charge C-rate	Discharge C-rate	Cycle Life	Calendar Life
Lead Acid (Flooded)	50%	80%	0.10C	0.20C	500 cycles	5 years
Lead Acid (AGM)	50%	85%	0.20C	0.30C	700 cycles	7 years
Lead Acid (Gel)	50%	85%	0.15C	0.25C	800 cycles	8 years
Lithium LFP	90%	95%	0.50C	1.00C	4000 cycles	15 years
Lithium NMC	80%	93%	0.50C	1.00C	2500 cycles	10 years

■ *DoD = Depth of Discharge. C-rate: 1C discharges the full capacity in 1 hour.*

11.2 Capacity & Energy Calculations

```
Nominal_Capacity_kWh = (V_nominal × Ah_per_unit × quantity) / 1000 Age_years =
days_since_installation / 365.25 Age_factor = max(0.70, 1.0 − Age_years ×
degradation_per_year_rate) [70% is the industry-standard replacement threshold]
Usable_Capacity_kWh = Nominal_kWh × DoD × Age_factor Available_Energy_kWh = Usable_kWh
× SOC_current [SOC ∈ 0..1] Max_Charge_Power_kW = Nominal_kWh × C_rate_charge
Max_Discharge_Power_kW = Nominal_kWh × C_rate_discharge Runtime_at_max_disch =
Usable_kWh / Max_Discharge_kW [hours] Runtime_at_avg_disch = Usable_kWh /
(Max_Discharge_kW/2) [hours]
```

11.3 Health Status Classification

Age Factor Range	Health Status	Interpretation
≥ 0.90	Good	Full rated performance, minimal degradation
0.80–0.90	Fair	Noticeable capacity loss, continue monitoring
0.70–0.80	Degraded	Significant capacity loss, plan replacement
< 0.70	Replace	Below industry threshold, immediate replacement

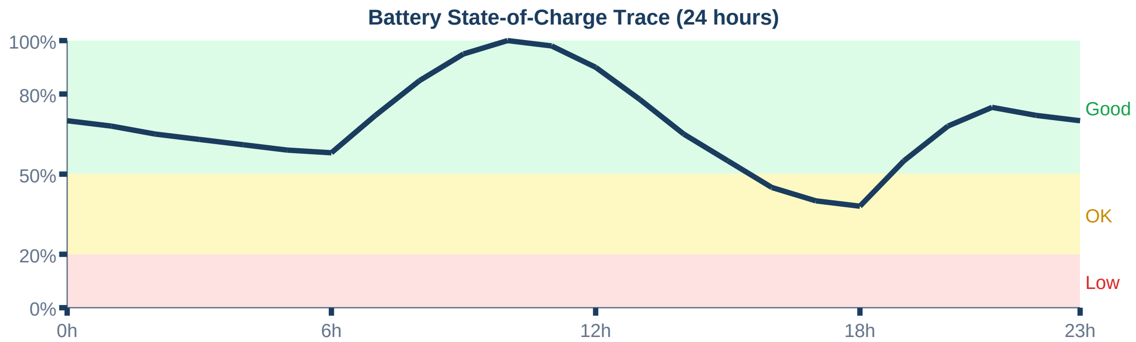


Figure 11.1 — Simulated 24-hour battery SOC trace with charging (morning solar) and discharge (evening)

11.4 Solar System – Battery Pairing

Each battery can be paired with a named solar system. Surplus solar energy from a paired system is directed preferentially to its paired batteries before the common pool. This allows modelling of multiple isolated DC buses.

12. Source Dispatch Algorithm

The dispatch algorithm runs hour by hour across a 24-hour window and determines, for each hour, how much power each source contributes to the load. The priority order follows economic and environmental logic: free solar first, then stored solar energy (batteries), then purchased grid power, and finally the most expensive and polluting option — diesel generation.

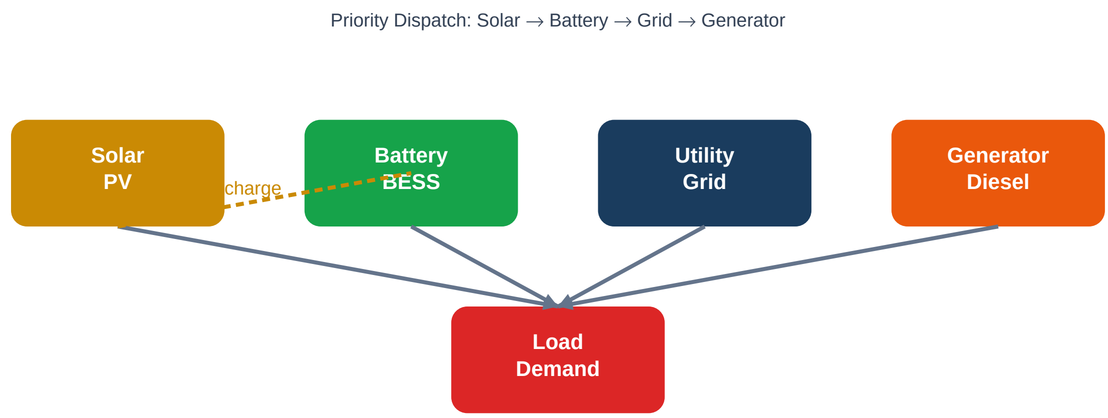


Figure 12.1 — Source dispatch priority diagram

12.1 Dispatch Priority Logic (per hour h)

```
Demand_h = Load_kW[h] (from load profile) Step 1: Solar covers load Solar_used[h] =
min(Solar_kW[h], Demand_h) Remaining_h = Demand_h − Solar_used[h] Surplus_solar =
Solar_kW[h] − Solar_used[h] Step 2: Solar charges batteries (paired first, then pool)
For each battery b paired to solar system s: Headroom_b = (Usable_kWh_b −
Current_kWh_b) × 1000 [W] Charge_b = min(Surplus/system, MaxChargeRate_b, Headroom_b /
ηcharge_b) SOC_b += Charge_b × η_roundtrip / Nominal_kWh_b / 1000 Step 3: Batteries
discharge to cover remaining load Ceil_W = min( ΣMaxDischargePower,
Σ(Current_kWh×1000), ΣInverterCap ) Each battery contributes proportionally to its SOC:
Batt_share_b = Ceil_W × (Current_kWh_b / Σ Current_kWh) Discharge_b = min(Batt_share_b,
MaxDischargePower_b, InverterCap_b) SOC_b -= Discharge_b / η_roundtrip / Nominal_kWh_b
Battery_discharged[h] = Σ Discharge_b / 1000 [kW] Remaining_h -= Battery_discharged[h]
Step 4: Utility grid Utility[h] = min(UtilityCapacity_kW, Remaining_h) Remaining_h -=
Utility[h] Step 5: Generator Generator[h] = min(GeneratorCapacity_kW, Remaining_h)
Remaining_h -= Generator[h] Step 6: Opportunistic generator battery charging if
Generator[h] > 0 and GeneratorCapacity_kW > 0: Gen_max_for_charge =
GeneratorCapacity_kW × 0.85 Spare_gen = max(0, Gen_max_for_charge − Generator[h]) For
each battery b: AC_charge_b = min(share, MaxChargeRate_b × 1000) DC_charge_b =
AC_charge_b × η_AC_DC × η_roundtrip SOC_b += DC_charge_b / 1000 / Nominal_kWh_b
Generator[h] += AC_charge_b / 1000 Unmet[h] = max(0, Remaining_h) (shortfall — capacity
insufficient)
```

12.2 Generator Efficiency Constants

Constant	Value	Meaning
GEN_OPTIMAL_MAX_LOAD	0.85 (85%)	Upper bound of efficient load band for diesel generators
INV_EFF (AC→DC)	0.95 (95%)	Efficiency of AC-to-DC conversion when charging from generator
Battery round-trip η	Chemistry-specific (80–95%)	DC charging/discharging efficiency

12.3 Output Statistics (per month/simulation run)

- solar_kwh — Total solar energy generated and used [kWh]
- solar_self_consumption % — Solar energy used directly or stored, not curtailed
- battery_charged_kwh — Total energy stored in batteries [kWh]
- battery_efficiency_loss_kwh — Round-trip losses in BESS [kWh]
- utility_kwh — Energy drawn from the grid [kWh]
- generator_kwh — Energy produced by diesel generator [kWh]
- generator_loading_avg % — Average loading of generator (efficiency proxy)
- unmet_kwh — Unserved energy demand [kWh] (capacity gap alert)
- final_soc — Battery SOC at end of 24-hour window [0..1]

13. 24-Hour Load Profile Generation

The load profile translates rated component powers and schedules into a time-resolved 24-hour demand curve. It enables dispatch simulation and visual insight into when loads are active.

13.1 Component Schedule Model

Each component carries three independent schedule dimensions:

Dimension	Options	Example
usage_season	all / summer / winter / spring / fall	summer (HVAC cooling)
usage_day_type	all / weekday / weekend	weekday (office hours)
usage_time_intervals	List of {start, end} pairs (HH:MM)	[{08:00,13:00},{14:00,18:00}]

13.2 Hourly Active Power Calculation

```
peak_W_i = S_i × PF_i × Quantity_i × DF_i For each hour h (0 – 23): P[h] = 0 for each
component i: if priority_i == 'critical': P[h] += peak_W_i / 1000 [always on, 24 hours]
else: for each interval [start, end] in usage_time_intervals_i: if start < end: #
normal interval if start ≤ h < end: P[h] += peak_W_i / 1000 else: # midnight-crossing
interval (e.g. 22:00–06:00) if h ≥ start or h < end: P[h] += peak_W_i / 1000
```

13.3 Hourly Reactive Power Calculation

```
For each hour h (0 – 23): Q[h] = 0 for each component i where PF_i < 1.0: ϕ_i =
arccos(PF_i) q_i = peak_W_i × tan(ϕ_i) / 1000 [kVAR] Add q_i to Q[h] for each hour
where i is active
```

13.4 JSON API Response Structure

```
{ "hourly_kw": [kW■, kW■, ..., kW■■], // 24 values "hourly_kvar": [kVAR■, ...,
kVAR■■], // 24 values "hourly_solar_kw": [Solar■, ..., Solar■■], // 24 values
"solar_data_source": "nasa_power" | "static_lookup", "components": [ { "peak_w": float,
"power_factor": float, "priority": "critical" | "essential" | "normal",
"usage_time_intervals": [{"start":"HH:MM","end":"HH:MM"}], "usage_season":
"all|summer|winter|spring|fall", "usage_day_type": "all|weekday|weekend" } ] }
```

14. Database Schema & Data Model

14.1 Entity Hierarchy Tables

Table	Key Columns	Foreign Keys
projects	id, user_id, name, building_type, location_lat, location_lng, solar_power, generator_power, work_days(JSON),	user_id → users
buildings	id, project_id, name, type, area	project_id → projects
floors	id, building_id, name, area	building_id → buildings
rooms	id, floor_id, type, name, area	floor_id → floors

14.2 Component Tables (4 levels)

Table	Unique Columns	Shared Columns
room_components	room_id	component_type_id, power(VA), phases, phase(A/B/C), power_factor, quantity, group
floor_components	floor_id	(same as above)
building_components	building_id	(same as above)
project_components	project_id	(same as above)

14.3 Power Source Tables (Polymorphic)

Utility lines, generator lines, and sockets use Laravel polymorphic relations. A single table covers all hierarchy levels via **sourceable_type** + **sourceable_id**.

Table	Polymorphic Columns	Other Columns
utility_lines	utilitylineable_type, utilitylineable_id	name, power(VA), phases
generator_lines	generatorlineable_type, generatorlineable_id	name, power(VA), phases
sockets	socketable_type, socketable_id	phase_type, quantity, power(always 200)

14.4 Battery & Solar Tables

Column	Type	Description
name	string	User-friendly label
chemistry	enum	lead_acid_flooded | agm | gel | lithium_lfp | lithium_nmc
nominal_voltage_v	float	Nominal bank voltage [V]
capacity_ah_per_unit	float	Capacity of each unit [Ah]
quantity	integer	Number of units in parallel
installation_date	date	Used to calculate age and degradation

Column	Type	Description
depth_of_discharge	float	Usable fraction of nominal capacity (0..1)
round_trip_efficiency	float	Charge + discharge efficiency combined (0..1)
c_rate_charge	float	Max charge rate as fraction of capacity [C]
c_rate_discharge	float	Max discharge rate [C]
current_soc	float	Current state of charge (0..1)
solar_system_id	FK|null	Paired solar system (nullable)
is_active	boolean	Include in dispatch simulation

15. REST API Reference

All 56 API endpoints are prefixed with **/api** and protected by Laravel Sanctum token authentication (except public endpoints noted). The API follows REST conventions with JSON request/response bodies.

Projects

Method	Endpoint	Description
GET	/projects	List all user projects
POST	/projects	Create new project
GET	/projects/{id}	Get project detail
PUT	/projects/{id}	Update project
DELETE	/projects/{id}	Delete project
GET	/projects/{id}/total-power	Calculate total power demand
GET	/projects/{id}/load-profile	Generate 24-h load profile
GET	/projects/{id}/phase-balance	Calculate 3-phase balance
GET	/projects/{id}/backup	Export full project backup

Buildings / Floors / Rooms

Method	Endpoint	Description
GET	/projects/{id}/buildings	List buildings
POST	/projects/{id}/buildings	Create building
PUT	/buildings/{id}	Update building
DELETE	/buildings/{id}	Delete building
GET	/buildings/{id}/total-power	Building power calculation
POST	/buildings/{id}/apply-optimal-phase	Apply optimal phase balance
GET	/buildings/{id}/floors	List floors
GET	/floors/{id}/rooms	List rooms

Batteries & Solar

Method	Endpoint	Description
GET	/projects/{id}/batteries	List batteries
POST	/projects/{id}/batteries	Add battery bank
GET	/batteries/{id}	Get battery detail with computed metrics
PUT	/batteries/{id}	Update battery
DELETE	/batteries/{id}	Remove battery

Method	Endpoint	Description
POST	/batteries/{id}/reset-soc	Reset SOC to specified value
POST	/batteries/{id}/runtime-at-load	Calculate runtime at given load
GET	/battery-chemistry-defaults	PUBLIC: Get chemistry presets
GET	/projects/{id}/solar-systems	List solar systems
POST	/projects/{id}/solar-systems	Add solar system
PUT	/solar-systems/{id}	Update solar system
DELETE	/solar-systems/{id}	Remove solar system

Power Sources (Polymorphic)

Method	Endpoint	Description
GET	/{entity}/{id}/utility-lines	List utility lines
POST	/{entity}/{id}/utility-lines	Add utility line
GET	/{entity}/{id}/generator-lines	List generator lines
POST	/{entity}/{id}/generator-lines	Add generator line
GET	/{entity}/{id}/sockets	List socket outlets
POST	/{entity}/{id}/sockets	Add socket outlets

Admin (Public login)

Method	Endpoint	Description
POST	/admin/login	PUBLIC: Admin credential login
GET	/admin/users	List all users (admin)
PUT	/admin/users/{id}	Edit user (admin)
DELETE	/admin/users/{id}	Delete user (admin)

16. Frontend Architecture & UI Logic

The frontend is a React 18 Single-Page Application (SPA) using React Router v6 for client-side navigation. State management is kept simple — component-local useState/useEffect with an AuthContext for global user session. All engineering results are rendered through Recharts charting components.

16.1 Page Inventory

Page Component	Route	Functionality
LoginPage	/login	Google OAuth 2.0 sign-in
AdminLoginPage	/admin/login	Credential-based admin login
DashboardPage	/dashboard	Projects list and quick stats
CreateProjectPage	/projects/new	New project wizard
ProjectPage	/projects/:id	Project overview, sources, members, batteries
BuildingsPage	/projects/:id/buildings	Building list with power summaries
BuildingPage	/buildings/:id	Floor list, components, reactive power panel
FloorPage	/floors/:id	Room list, floor components
RoomDetailPage	/rooms/:id	Room components and sockets editor
LoadSchedulePage	/projects/:id/schedule	24-h load profile, dispatch charts, SOC trace
PhaseBalancePage	/projects/:id/phase-balance	3-phase phasor display, optimal assignment
AdminDashboardPage	/admin	User management table

16.2 LoadSchedulePage — Key UI Sections

This is the most complex page (1042 lines). It fetches both the load profile and the dispatch schedule, then renders four tabbed views:

Tab	Charts / Widgets Shown	Data Source API
Load Profile	24-h area chart: load demand vs solar generation	/load-profile
Sources	Stacked bar chart per source (Solar/Battery/Grid/Generator)	/schedule
Combined Dispatch	ComposedChart overlaying all sources + demand line	/schedule
Battery SOC	Line chart of SOC trace + charge/discharge events	/schedule

16.3 Key UI Components

Component	Purpose
PowerBanner	Displays S (VA), P (kW), Q (kVAR), PF, I (A) for any entity level
ReactivePowerPanel	Shows kVAR breakdown, PF status, capacitor bank recommendation

Component	Purpose
EntityComponents	Reusable form: add/edit/delete components with all fields
EntityScheduleModal	Configure work hours, day types, season intervals per entity
TimeScheduleModal	Manage time intervals with add/remove controls
PowerSourcesBanner	Cards for each active source: utility, generator, solar capacity
ProjectSidebar	Hierarchical tree: buildings → floors → rooms with links
BackupChoiceModal	Select backup scope (project/building/floor/room) for export
ProjectMembersModal	Add users by email, set roles (admin/main/normal)

16.4 Recharts Configuration

All charts use **ResponsiveContainer width='100%'** so they adapt to any screen width. The primary chart type for the load profile is **ComposedChart** with **Area** (filled, for solar) overlaid with **Line** (for total load). The dispatch view uses a **StackedBarChart** with one bar segment per source type.

17. Key Formulae Quick Reference

Formula	Equation	Units
Apparent Power	$S = \sqrt{(P^2 + Q^2)}$	VA
Active Power	P = S × cos(ϕ)	W
Reactive Power	Q = P × tan(arccos(PF))	VAR
Power Factor	PF = P / S = cos(ϕ)	—
3-phase line current	I = S / (√3 × 400)	A
1-phase line current	I = S / 230	A
Room component DF	DF = DF_room × DF_f2b × DF_r2f × 0.70	—
Socket demand (n≤10)	D = n × 200	VA
Socket demand (10<n≤20)	D = 2000 + (n-10)×150	VA
Socket demand (n>20)	D = 3500 + (n-20)×80	VA
Q_cap needed	Q_c = P×tan(arccos(PF_sys)) – P×tan(arccos(0.95))	VAR
Capacitor bank (rounded)	Q_bank = ■Q_c/500■×500	VAR
Delta capacitance	$C = Q\_bank/(3 \times 2\pi \times 50 \times 400^2)$	F
Motor inrush add	ΔVA = 0.25 × VA_largest_motor	VA
Solar capacity	P_cap = Area × 0.17 × 1000 × 0.75	W
Solar declination	δ = 23.45 × sin(360/365×(doy–81))	°
Hour angle sunrise	cos(H) = –tan(ϕ)×tan(δ)	—
Sunrise time	t_rise = 12 – arccos(cos(H))/15	h
Solar output (hourly)	P[h] = P_peak × sin(π×(mid–rise)/daylight)	W
Battery nominal capacity	E_nom = V × Ah × n / 1000	kWh
Battery age factor	f_age = max(0.70, 1–age_yr×degr_rate)	—
Battery usable capacity	E_use = E_nom × DoD × f_age	kWh
Max discharge power	P_disc = E_nom × C_rate_discharge	kW
Inverter AC→DC eff	η_AC_DC = 0.95	—
Phase imbalance %	Imb = (I_max–I_min)/I_avg × 100	%

18. Summary & Conclusion

Power Profile demonstrates the successful integration of rigorous electrical engineering standards with modern web technologies to produce a practical, accurate, and user-friendly power system design tool.

18.1 Achievements

- Implemented IEC 60364-8-1 diversity factors with full building-type and room-type coverage across a 4-level entity hierarchy.
- Tiered socket demand model consistent with IEC 60364-5-52 demand factor schedules.
- Complete AC power triangle vector model — active (W), reactive (VAR), and apparent (VA) — with proper phasor aggregation.
- PENRA-compliant power factor correction with delta capacitor bank sizing to the nearest 0.5 kVAR step.
- NEC Article 430 / IEC 60947-4 motor inrush (125%) applied to maximum demand vectors.
- Solar PV model using Spencer's declination, hour-angle sunrise/sunset, and sinusoidal bell profile — calibrated by real NASA POWER satellite GHI data.
- Multi-chemistry BESS model with age-based degradation, SOC tracking, and C-rate constraints.
- Greedy hour-by-hour multi-source dispatch (Solar → BESS → Grid → Generator) with opportunistic generator charging.
- 3-phase load balancing with greedy First-Fit Decreasing phase assignment algorithm.
- RESTful Laravel 13 API (56 endpoints, Sanctum auth, Google OAuth) consumed by React 18 SPA.
- Interactive 24-hour charts (Recharts) showing load profile, solar generation, dispatch breakdown, and SOC trace.

18.2 Standards Compliance Summary

Standard	Feature	Compliance
IEC 60364-8-1	Diversity factors	✓ Full table implemented
BS 7671	Diversity factor source	✓ Cross-referenced
CIBSE Guide C	Room coincidence factors	✓ All room types covered
PENRA	PF = 0.95 target	✓ Automatic correction sizing
NEC Art. 430	Motor 125% rule	✓ Applied to all motor loads
IEC 60947-4	Motor inrush	✓ Consistent with NEC
IEC 60831	Capacitor delta config	✓ Δ sizing equation implemented
NASA POWER v2	GHI irradiance	✓ Real satellite API + caching

Power Profile — Graduation Project Report
Author: Ahmed Zoher | June 2026
Stack: Laravel 13 / PHP 8.4 + React 18 / Vite + SQLite + NASA POWER
Standards: IEC 60364-8-1 · BS 7671 · PENRA · NEC · CIBSE